

Mathematical Modelling- Hybrid Epidemic Deterministic and Stochastic-Hospital Model

MATH F420 GROUP PROJECT

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Abstract

In this paper, we are analysing a deterministic SIHR (Susceptible, Infected, Hospitalized, Recovered) mathematical model obtained using stochastic methodologies. To capture realistic healthcare utilization, the model initializes hospitalized populations using a Continuous-Time Markov Chain (CTMC) before transitioning into a deterministic Ordinary Differential Equation (ODE) framework. An important feature of this paper is the integration of behavioural feedback mechanism and sensitivity of disease to hospitalizations where the transmission rate ($\beta(H)$) is mathematically dependent on public awareness and hospital congestion relative to the total healthcare capacity (C). The Basic Reproduction number will be computed using the Next Generation Matrix Method and phase portrait analysis is also conducted. Further, Markov Chain Monte Carlo (MCMC) simulations are used in order to reverse-engineer the baseline transmission rates from empirical real-life hospital admission data(Covid-19). The findings establish analytical thresholds for policy intervention which demonstrate the exact hospital load that will be required to trigger public response and suppress transmission. Ultimately, this framework provides insight on how hospital capacity and policy intervention can prevent healthcare collapse in an epidemic.

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1 Introduction

1.1 Working of Hospital Systems during an epidemic fallout

Hospital and healthcare facility needs, similar to other vital institutions, rely on the effective coordination of capacity and individuals accessing the service during a specific time frame. In the event of an epidemic, owing to scarce resources and an unprecedented surge in patients, it becomes crucial to reduce the burden on hospitals and avoid additional negligent interactions. This requires modeling the outbreak scenario while considering the present hospital capacity and any medical resources designated for that time. This mathematical modelling project focuses on developing a suitable framework by incorporating aspects of stochastic queuing systems into the current epidemiological models to formulate a more appropriate and realistic mathematical representation of the situation.

1.2 Problem Statement

Consider a Hospital ABC operational during a pandemic. It has been operational with adaptive capacity and requires a policy intervention based on epidemiological modelling.

1. Develop a hybrid deterministic–stochastic framework where the epidemic dynamics are modeled using a system of nonlinear ordinary differential equations, while hospital service dynamics are represented through by ODE Formulation by CTMC initialization.
2. Investigate how hospital congestion influences epidemic transmission by incorporating a feedback mechanism in which the transmission or recovery rates depend on the level of hospital utilization.
3. Analyze the impact of varying hospital capacity on the stability of the epidemic dynamics and determine the conditions under which healthcare systems remain resilient to infection surges.
4. Construct the phase portraits for the interacting populations, namely the Hospitalized and Infected compartments. Perform a linear stability analysis of the equilibrium points to determine the long-term behavior of the system.
5. Provide actionable insights on the hospital capacity and public awareness factor required to prevent healthcare collapse and stabilize the dynamics.

1.3 CTMC-Based Stochastic Initialization and Deterministic PDE Formulation for Hospitalized Individuals

Let $X(t)$ be the number of hospitalized individuals.

Model $X(t)$ as a continuous-time Markov chain with two types of jumps:

- **Admission:** $X \rightarrow X + 1$ at rate $\lambda(t) = pI(t)$
- **Discharge:** $X \rightarrow X - 1$ at rate $\mu(X) = uX$

This is a **time-inhomogeneous birth–death process** because $I(t)$ changes over time. The generator function for the particular chain would given by-

$$(\mathcal{L}_t f)(x) = pI(t) [f(x+1) - f(x)] + ux [f(x-1) - f(x)].$$

The generator encodes all transition rates and determines how the continuous-time Markov chain changes over an infinitesimal time interval. It is also required because we use the expected values of $X(t)$ in the partial differential equation that we derive from the CTMC, and that derivation is done by using the generator(as done further).

By Dynkin's formula,

$$\frac{d}{dt} \mathbb{E}[f(X(t))] = \mathbb{E}[(\mathcal{L}_t f)(X(t))].$$

Choosing $f(x) = x$, we obtain

$$(\mathcal{L}_t f)(x) = pI(t) - ux.$$

Thus,

$$\frac{d}{dt}\mathbb{E}[X(t)] = pI(t) - u\mathbb{E}[X(t)].$$

Defining $H(t) = \mathbb{E}[X(t)]$, we obtain the deterministic equation

$$\frac{dH}{dt} = pI - uH.$$

Remark 1. *The function $H(t)$ denotes the average number of hospitalized individuals at time t , computed across all possible realizations of the stochastic process. Since the probability distribution of hospitalizations evolves as the epidemic progresses, this average is itself time-dependent. Thus, $H(t)$ provides a smooth, deterministic approximation of the stochastic variable $X(t)$ at a macroscopic level.*

Even in basic epidemiological models with Susceptible, Infected and Recovery Compartments, S, I, R variables usually denote the expected values rather than the exact number of individuals.

2 Main Working of the Project

The dynamics of the system are given by the following set of differential equations:

$$\frac{dS}{dt} = \Lambda - \beta(H)SI - \mu S \quad (1)$$

$$\frac{dI}{dt} = \beta(H)SI - \gamma I - pI - \mu I \quad (2)$$

$$\frac{dH}{dt} = pI - uH - \mu H \quad (3)$$

$$\frac{dR}{dt} = \gamma I + uH - \mu R \quad (4)$$

The transmission rate is defined as:

$$\beta(H) = \beta_0 \left[1 + \alpha \left(\frac{H}{C} - 1 \right)^2 - \alpha \right] \quad (5)$$

Description of Variables and Parameters

- $S(t)$: Number of susceptible individuals at time t
- $I(t)$: Number of infected individuals at time t
- $H(t)$: Number of hospitalized individuals at time t
- $\beta(H)$: Transmission rate, dependent on the hospitalized population
- β_0 : Baseline transmission rate
- γ : Recovery rate of infected individuals
- p : Rate at which infected individuals become hospitalized
- u : Rate at which hospitalized individuals leave the hospital (recovery)
- α : Strength of the effect of hospitalization on transmission
- C : Scaling constant (e.g., healthcare capacity)
- Λ : Birth
- μ : Death Rate

In this model, susceptible individuals become infected at a rate proportional to $\beta(H)SI$. Infected individuals either recover at rate γ or are hospitalized at rate p . Hospitalized individuals leave the hospital at rate u . The transmission rate $\beta(H)$ depends on the number of hospitalized individuals, introducing a feedback mechanism into the system. The assumption is that the population of the system is constant. That would mean-

Let the total population be defined as:

$$N = S + I + H + R \quad (6)$$

Differentiating with respect to time:

$$\frac{dN}{dt} = \frac{dS}{dt} + \frac{dI}{dt} + \frac{dH}{dt} + \frac{dR}{dt} \quad (7)$$

Substituting from the model equations:

$$\frac{dN}{dt} = \Lambda - \mu N \quad (8)$$

For a constant population, we require:

$$\frac{dN}{dt} = 0 \quad (9)$$

Thus, the constant population condition is:

$$\Lambda = \mu N \quad (10)$$

This will help us to reduce this model into a 3-compartment model for further analysis.

2.1 Calculation of equilibrium points

Assuming a constant total population ($N = \Lambda/\mu$), the system is reduced to three compartments: Susceptible (S), Infected (I), and Hospitalized (H). The dynamics are governed by the following set of differential equations:

$$\frac{dS}{dt} = \Lambda - \beta(H)SI - \mu S \quad (11)$$

$$\frac{dI}{dt} = \beta(H)SI - (\gamma + p + \mu)I \quad (12)$$

$$\frac{dH}{dt} = pI - (u + \mu)H \quad (13)$$

The transmission rate, $\beta(H)$, incorporates a behavioral feedback mechanism where public awareness reduces transmission as hospitalizations approach the healthcare capacity C . The corrected formulation is:

$$\beta(H) = \beta_0 \left[1 + \alpha \left(\frac{H}{C} - 1 \right)^2 - \alpha \right] \quad (14)$$

For algebraic convenience during the equilibrium analysis, we expand Equation 14:

$$\beta(H) = \beta_0 \left(1 - \frac{2\alpha}{C}H + \frac{\alpha}{C^2}H^2 \right) \quad (15)$$

Equilibrium points occur when the rates of change for all compartments are zero ($\frac{dS}{dt} = \frac{dI}{dt} = \frac{dH}{dt} = 0$). From Equation 13:

$$pI - (u + \mu)H = 0 \implies I = \frac{u + \mu}{p}H \quad (16)$$

From Equation 12:

$$I[\beta(H)S - (\gamma + p + \mu)] = 0 \quad (17)$$

This yields two distinct equilibrium states:

Disease-Free Equilibrium (DFE)

If the disease is eradicated, $I = 0$. By Equation 16, if $I = 0$, then $H = 0$. Substituting $I = 0$ into Equation 11 yields:

$$\Lambda - 0 - \mu S = 0 \implies S = \frac{\Lambda}{\mu} \quad (18)$$

Thus, the Disease-Free Equilibrium is:

$$E_0 = (S_0, I_0, H_0) = \left(\frac{\Lambda}{\mu}, 0, 0 \right) \quad (19)$$

Endemic Equilibrium (EE)

If the disease persists, $I \neq 0$. Therefore, the bracketed term in the infected equation must be zero:

$$\beta(H^*)S^* - (\gamma + p + \mu) = 0 \implies S^* = \frac{\gamma + p + \mu}{\beta(H^*)} \quad (20)$$

We substitute I^* (from Eq 16) and S^* (from Eq 20) into the susceptible equation (Eq 11):

$$\Lambda - \beta(H^*) \left(\frac{\gamma + p + \mu}{\beta(H^*)} \right) \left(\frac{u + \mu}{p} H^* \right) - \mu \left(\frac{\gamma + p + \mu}{\beta(H^*)} \right) = 0 \quad (21)$$

Notice that $\beta(H^*)$ cancels out in the middle term. To simplify, let K be the aggregate transition constant:

$$K = \frac{(\gamma + p + \mu)(u + \mu)}{p} \quad (22)$$

The equation simplifies to:

$$\Lambda - KH^* - \frac{\mu(\gamma + p + \mu)}{\beta(H^*)} = 0 \quad (23)$$

Rearranging to isolate $\beta(H^*)$:

$$\beta(H^*)(\Lambda - KH^*) = \mu(\gamma + p + \mu) \quad (24)$$

Substituting the expanded transmission function (Eq 15):

$$\beta_0 \left(1 - \frac{2\alpha}{C}H^* + \frac{\alpha}{C^2}H^{*2} \right) (\Lambda - KH^*) = \mu(\gamma + p + \mu) \quad (25)$$

Dividing both sides by β_0 and expanding the left-hand side yields a cubic polynomial in terms of H^* :

$$-\left(\frac{\alpha K}{C^2}\right)H^{*3} + \left(\frac{\alpha\Lambda}{C^2} + \frac{2\alpha K}{C}\right)H^{*2} - \left(\frac{2\alpha\Lambda}{C} + K\right)H^* + \left(\Lambda - \frac{\mu(\gamma + p + \mu)}{\beta_0}\right) = 0 \quad (26)$$

The Endemic Equilibrium exists if there is a positive real root H^* to this cubic equation that satisfies $H^* < \Lambda/K$ (ensuring S^* remains positive).

Remark 2. To find the Endemic Equilibrium (S^*, I^*, H^*):

1. To find positive real roots of this cubic equation for H^* . For the equilibrium to be biologically valid, you must also check that $H^* < \frac{\Lambda}{K}$ (otherwise, your susceptible population S^* becomes negative).
2. Once you have H^* , you calculate $I^* = \frac{u+\mu}{p}H^*$.
3. Then you calculate $S^* = \frac{\gamma+p+\mu}{\beta(H^*)}$.

2.2 Reproduction number of the dynamic system using Next Generation Matrix method

To calculate the basic reproduction number (R_0) using the Next Generation Matrix (NGM) method, we focus exclusively on the disease compartments: Infected (I) and Hospitalized (H). Let the state vector be $\mathbf{x} = [I, H]^T$.

New Infections and Transitions

We partition the rates of change for the disease compartments into two vectors: new infections ($\mathcal{F}$) and transitions/removals ($\mathcal{V}$). From the system of differential equations, we have:

$$\mathcal{F} = \begin{bmatrix} \beta(H)SI \\ 0 \end{bmatrix}, \quad \mathcal{V} = \begin{bmatrix} (\gamma + p + \mu)I \\ -pI + (u + \mu)H \end{bmatrix} \quad (27)$$

Note that the term pI represents individuals transitioning from the infected compartment to the hospitalized compartment. Because this is a transfer between disease states rather than a brand-new infection from a susceptible individual, it is placed in the $\mathcal{V}$ matrix as a net flow.

Evaluation at Disease-Free Equilibrium (DFE)

The DFE occurs when there is no disease in the population, meaning $I = 0$ and $H = 0$. Substituting these values into the susceptible equation gives $S_0 = \frac{\Lambda}{\mu}$. Thus, the DFE is $E_0 = \left(\frac{\Lambda}{\mu}, 0, 0\right)$.

Crucially, we evaluate the updated transmission rate function at the DFE ($H = 0$):

$$\beta(0) = \beta_0 \left[1 + \alpha \left(\frac{0}{C} - 1 \right)^2 - \alpha \right] = \beta_0(1 + \alpha - \alpha) = \beta_0 \quad (28)$$

Jacobian Matrices F and V

We compute the local linearizations of $\mathcal{F}$ and $\mathcal{V}$ by taking their Jacobian matrices with respect to the disease variables (I, H) and evaluating them at the DFE (E_0).

The new infections matrix F is:

$$F = \begin{bmatrix} \frac{\partial \mathcal{F}_1}{\partial I} & \frac{\partial \mathcal{F}_1}{\partial H} \\ \frac{\partial \mathcal{F}_2}{\partial I} & \frac{\partial \mathcal{F}_2}{\partial H} \end{bmatrix}_{E_0} = \begin{bmatrix} \beta(0)S_0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} \frac{\beta_0 \Lambda}{\mu} & 0 \\ 0 & 0 \end{bmatrix} \quad (29)$$

The transitions matrix V is:

$$V = \begin{bmatrix} \frac{\partial \mathcal{V}_1}{\partial I} & \frac{\partial \mathcal{V}_1}{\partial H} \\ \frac{\partial \mathcal{V}_2}{\partial I} & \frac{\partial \mathcal{V}_2}{\partial H} \end{bmatrix}_{E_0} = \begin{bmatrix} \gamma + p + \mu & 0 \\ -p & u + \mu \end{bmatrix} \quad (30)$$

The Next Generation Matrix and R_0

To find the Next Generation Matrix, we first determine the inverse of the transition matrix, V^{-1} :

$$V^{-1} = \frac{1}{(\gamma + p + \mu)(u + \mu) - (0)(-p)} \begin{bmatrix} u + \mu & 0 \\ p & \gamma + p + \mu \end{bmatrix} = \begin{bmatrix} \frac{1}{\gamma + p + \mu} & 0 \\ \frac{p}{(\gamma + p + \mu)(u + \mu)} & \frac{1}{u + \mu} \end{bmatrix} \quad (31)$$

The Next Generation Matrix is given by the product $K = FV^{-1}$:

$$K = \begin{bmatrix} \frac{\beta_0 \Lambda}{\mu} & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\gamma + p + \mu} & 0 \\ \frac{p}{(\gamma + p + \mu)(u + \mu)} & \frac{1}{u + \mu} \end{bmatrix} = \begin{bmatrix} \frac{\beta_0 \Lambda}{\mu(\gamma + p + \mu)} & 0 \\ 0 & 0 \end{bmatrix} \quad (32)$$

The basic reproduction number, R_0 , is defined as the spectral radius (the dominant eigenvalue) of the Next Generation Matrix, denoted as $\rho(FV^{-1})$. Because K is a diagonal matrix, its eigenvalues lie on the main diagonal. Therefore, the basic reproduction number for the system is:

$$R_0 = \frac{\beta_0 \Lambda}{\mu(\gamma + p + \mu)} \quad (33)$$

2.3 Phase Portrait Analysis- I vs H

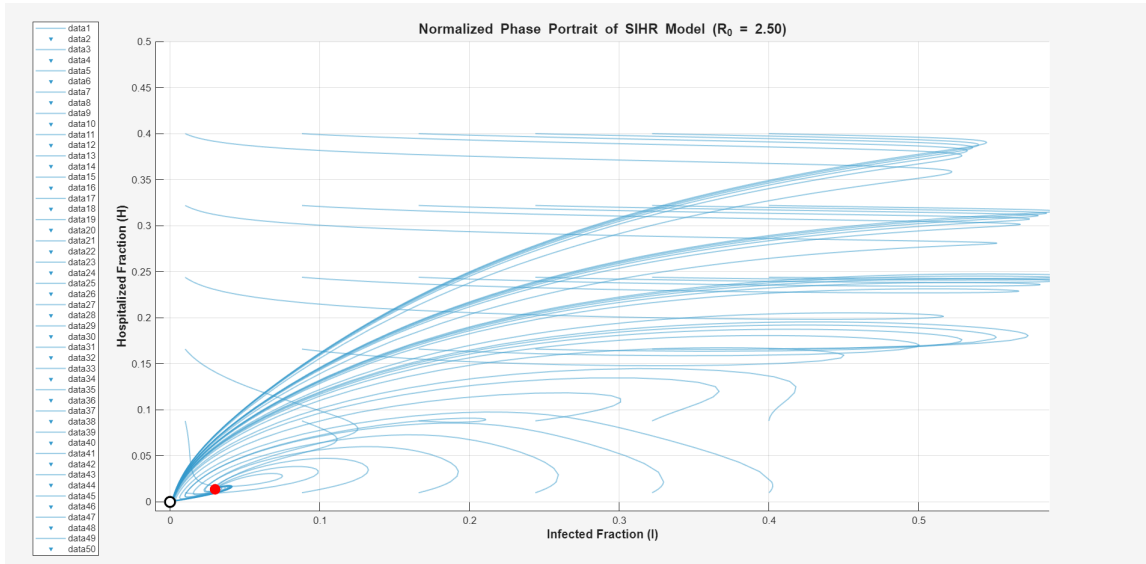


Figure 1: Global stability analysis at $Ro > 1$

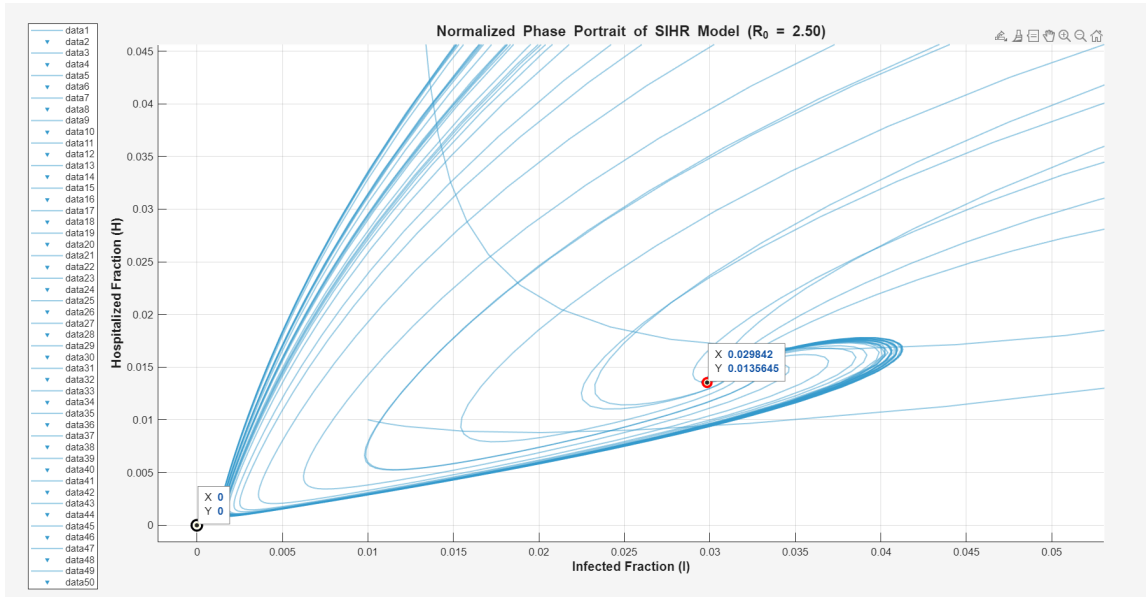


Figure 2: Local stability analysis at $R_0 > 1$

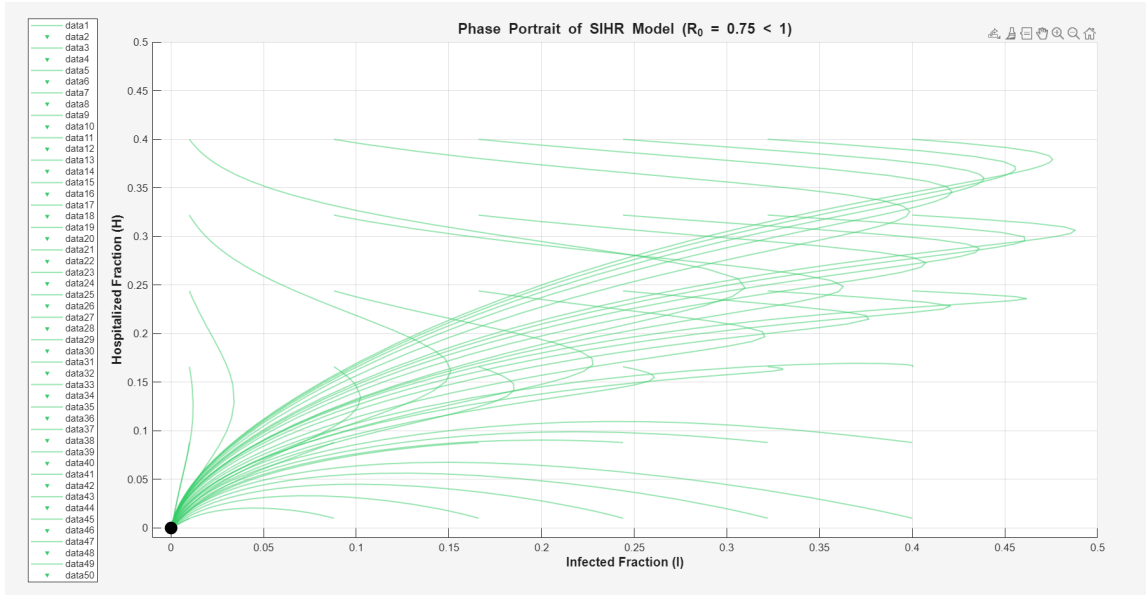


Figure 3: global stability analysis $R_0 < 1$

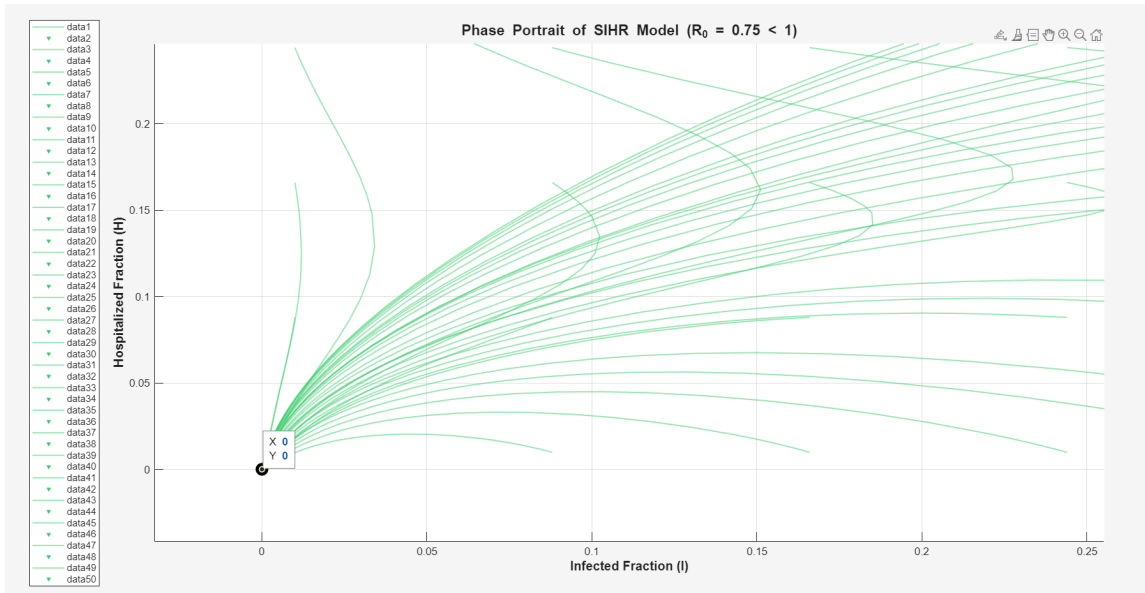


Figure 4: local stability $R_0 < 1$

There are certain graphs pertaining to Global stability of the system Even as an approximation, these portraits do perfectly illustrates a concept of Global Stability. It proves mathematically that no matter how bad the initial outbreak is (high I, high H), or how small it is, the system will always inevitably stabilize at that exact Endemic Equilibrium (the red dot) or the disease free equilibrium depending on the Reproduction number.

2.4 Effect of alpha on transmission of disease.

The core narrative of this SIHR model centers on the critical role of behavioral feedback, represented by the parameter α , in mitigating an epidemic. To analyze the effect of public awareness and hospitalization feedback on an outbreak, we can compare two extreme scenarios:

- $\alpha = 0.0$ (**No Feedback**): The public ignores rising hospitalizations. The transmission rate stays constant at β_0 . This represents a baseline “worst-case” scenario.
- $\alpha = 0.75$ (**Strong Feedback**): The public reacts strongly to overwhelmed hospitals (e.g., via social distancing, masks, or lockdowns). Transmission drops by up to 75% as hospitalizations approach healthcare capacity C .

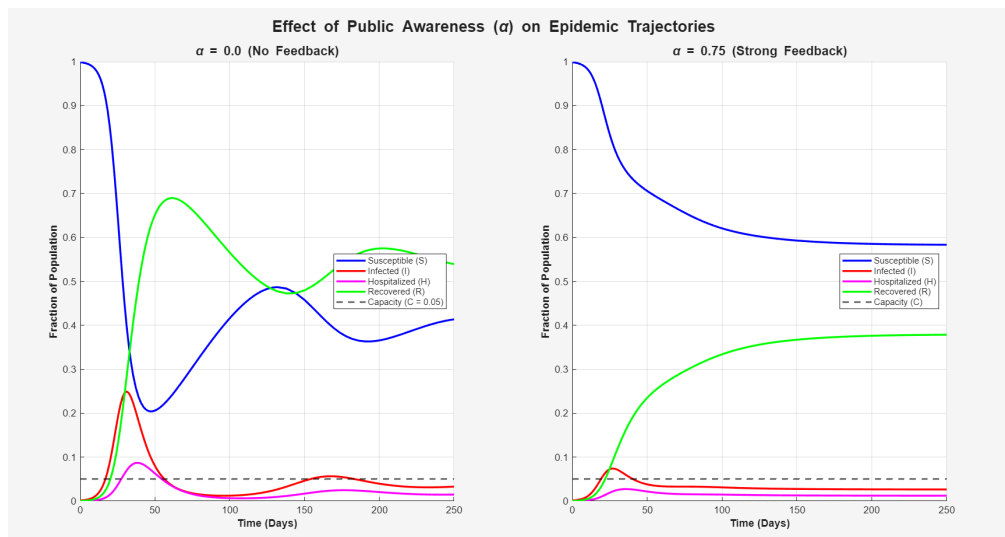


Figure 5: effect of alpha on disease transmission- balance of pathogen strength and public reaction

To further analyze the effect of public awareness and hospitalization feedback, we utilize the following specific transmission function:

$$\beta(H) = \beta_0 \left[1 + \alpha \left(\frac{H}{C} - 1 \right)^2 - \alpha \right] \quad (34)$$

2.5 Analytical Bounds and Suppression Thresholds

To quantify the impact of hospitalization levels on disease spread, we can calculate the absolute mathematical minimum for transmission within this framework. Assuming maximum public panic when hospitals are at or above capacity ($H \geq C$), the term $\left(\frac{H}{C} - 1\right)^2$ becomes zero. The equation simplifies to:

$$\beta_{min} = \beta_0(1 - \alpha) \quad (35)$$

Using baseline parameters for a strong feedback scenario (high sensitivity of hospitalization on transmission and highly infectious disease) ($\beta_0 = 0.6$ and $\alpha = 0.75$), we can check that the absolute minimum transmission rate drops to **0.15**.

To determine the specific hospital load required to achieve targeted reductions in the transmission rate, we evaluate the simplified effective transmission equation:

$$\beta(H) = 0.15 + 0.45 \left(\frac{H}{C} - 1 \right)^2 \quad (36)$$

2.5.1 Target 1: 50% Reduction in Transmission ($\beta \leq 0.30$)

To halve the baseline transmission (dropping it from 0.6 to 0.3), we solve for the ratio of hospitalizations to capacity (H/C):

$$\begin{aligned} 0.3 &= 0.15 + 0.45 \left(\frac{H}{C} - 1 \right)^2 \\ \frac{1}{3} &= \left(\frac{H}{C} - 1 \right)^2 \\ \frac{H}{C} &= 1 - \sqrt{\frac{1}{3}} \approx 0.423 \end{aligned}$$

Result: The hospitalized population must consistently remain greater than **42% of capacity** ($H > 0.42C$) to sustain a 50% reduction in the transmission rate.

2.5.2 Target 2: Extreme Suppression ($\beta \leq 0.20$)

To achieve a drastic drop in transmission, pushing it close to the absolute minimum of 0.15, the required threshold is calculated as:

$$\begin{aligned} 0.2 &= 0.15 + 0.45 \left(\frac{H}{C} - 1 \right)^2 \\ \frac{1}{9} &= \left(\frac{H}{C} - 1 \right)^2 \\ \frac{H}{C} &= 1 - \frac{1}{3} \approx 0.667 \end{aligned}$$

Result: Hospitalizations must exceed **67% of capacity** ($H > 0.67C$) to keep the epidemic under extreme suppression.

2.6 Hospitalization Equilibrium and Capacity Dynamics

To maintain a low transmission rate over an extended time domain (resulting in a "plateau" effect), the policy formation for the hospitalized to be in an optimized interval (calculated from subsection 2.5) for minimum β_H is essential to be done in time for any substantial increase in hospitalization. Based on the system of differential equations, the rate of change of the hospitalized population is governed by:

$$\frac{dH}{dt} = pI - uH - \mu H \quad (37)$$

At an epidemic peak, the rate of change is zero ($\frac{dH}{dt} = 0$). Setting the equation to zero yields the equilibrium relationship:

$$pI = (u + \mu)H \quad (38)$$

$$H = \left(\frac{p}{u + \mu} \right) I \quad (39)$$

Given that the natural death rate (μ) is generally negligible over the short duration of an acute epidemic, this can be approximated as:

$$H \approx \left(\frac{p}{u} \right) I \quad (40)$$

This relationship reveals the fundamental mechanics controlling the hospital load:

1. **Direct Proportionality:** The number of occupied hospital beds (H) is directly proportional to the hospitalization rate (p) multiplied by the active infectious population (I).
2. **Inflow vs. Outflow Balance:** The parameter p acts as the inflow variable from the infected pool, while u (the discharge/recovery rate) governs the outflow. A high p coupled with a low u will cause H to balloon rapidly, even under moderate infection levels.
3. **Triggering the Feedback Loop:** The hospitalization rate p is the crucial link back to the $\beta(H)$ transmission function. If p increases (e.g., due to a more severe pathogen strain), H scales up much faster, approaching the capacity limit C earlier in the timeline. Consequently, the strong public feedback mechanism is triggered almost immediately deciding the number of people to be hospitalized and the number to be remaining at home for treatment)

Ultimately, while the capacity C acts as the systemic ceiling that triggers intervention, p serves as the accelerator dictating how rapidly the epidemic trajectory hits that ceiling.

3 Monte Carlo simulation (approximation of best β_0)

The primary reason for executing this computational phase is to anchor the theoretical SIHR (Susceptible, Infected, Hospitalized, Recovered) model to empirical reality. While differential equations describe the mechanics of transmission, they require highly accurate initial conditions (I_0) and baseline transmission rates (β_0) to generate valid predictions. Because pathogens do not broadcast their transmission rates, β_0 cannot be measured directly; it must be inferred. To achieve this, Markov Chain Monte Carlo (MCMC) simulations are utilized to reverse-engineer these parameters from observed, noisy CSV data of daily hospital admissions. Depending on the time horizon being analyzed, the data requires two distinct modeling approaches.

3.1 2. The 60-Day Model: Short-Term Dynamics

For the short-term computational model fitted to the CSV data, we assume a closed population by explicitly removing birth and natural death rates ($\mu = 0$).

- **The Demographic vs. Viral Reality:** Human vital rates are incredibly slow. Assuming an average lifespan of 70 years, the daily vital rate is $\mu \approx 0.000039$. Over a 60-day window in a population of 10 million, only a marginal number of individuals are born or die naturally. In stark contrast, a novel virus grows exponentially over those same 60 days, potentially infecting hundreds of thousands of people.
- **Accuracy and Computational Efficiency:** The slow trickle of births has zero impact on the height or timing of the initial pandemic peak. Including μ implies that demographic changes are influencing the outbreak, which is factually incorrect for an acute 2-month wave. Furthermore, when fitting the curve to CSV data using MCMC, adding μ forces the algorithm to optimize a parameter that has no leverage on the shape of the curve, creating unnecessary computational noise.
- **The Short-Term Goal:** By setting $\mu = 0$, we apply Occam’s Razor—utilizing the simplest model that accurately describes the data. This results in a much cleaner, more precise fit for the variables that actually govern the transient dynamics, such as β_0 . Ultimately, this allows us to understand how high the initial hospital surge will be and whether it will exceed healthcare capacity (C). As demonstrated by the fitted MCMC data, this closed-population approach successfully tracks the initial outbreak, yielding an implied basic reproduction number of $R_0 = 2.79$ over the 60-day period.

3. The Long-Term Model: Asymptotic Analysis

Conversely, for the theoretical, asymptotic model as time approaches infinity ($t \rightarrow \infty$), we assume an open population and include vital dynamics ($\mu > 0$).

- **Mathematical Necessity and Endemic Reality:** In a closed population with no new births, the Susceptible (S) compartment strictly decreases. Eventually, the virus runs out of “fuel,” and the mathematical system is forced into a Disease-Free Equilibrium. In the real world, however, diseases do not simply disappear; they become endemic. Over a span of years, a constant influx of new susceptible individuals is introduced into the system via birth.
- **Unlocking Equilibria and Bifurcations:** Adding the birth rate (μN) mathematically balances the rate of infection, preventing S from dropping to zero. This allows the system to settle into a stable, non-zero Endemic Equilibrium point (S^*, I^*, H^*). Moreover, without a mechanism that constantly replenishes the susceptible population, it is impossible to study complex long-term behaviors like Hopf bifurcations or oscillating infection waves.
- **The Long-Term Goal:** The inclusion of vital dynamics is essential to understand the asymptotic stability of the system. This allows us to model what the permanent, baseline load on the hospital system will be years after the initial shock has passed.

Best beta: 4.179e-08
Best I0: 693
Implied R0: 2.79 <- should be 1.5 to 3.5

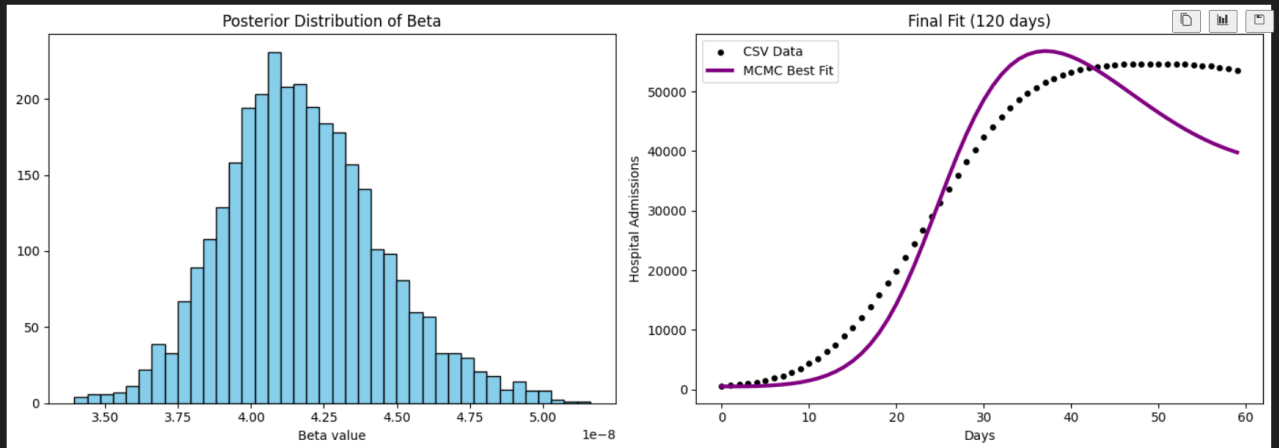


Figure 6: $R_0 = 2.79$ with initial days of outbreak

4 Conclusion

The Hybrid SIHR Model described in this study shows that there are substantial differences between theoretical models of epidemic spread and the dynamics of hospital services. The mathematical modeling of the research shows that, due to the raised awareness and attention brought about by the rise in hospitalizations, that feedback from the public can help reduce the spread of disease.

Based on the analysis discussed within this study, several key conclusions can be drawn:

- **Feedback Mitigates Transmission:** The reduction in the magnitude of transmission can be significantly increased by a high level of public response ($\alpha = 0.75$) when hospitalizations approach their maximum capacity when compared to a zero-feedback baseline scenario.
- **Actionable Suppression Thresholds:** Analytical bounds reveal that to achieve a 50% reduction in the baseline transmission rate, hospitalizations must consistently exceed 42% of total capacity. Achieving extreme suppression requires hospitalizations to surpass 67% of capacity.
- **System Stability:** The Stability of the System: The phase portrait analysis shows that the system is globally stable and that the trajectory of the outbreak will end at either a Disease-Free Equilibrium point or an Endemic Equilibrium point, depending on the value of the basic reproduction number (R_0).
- **Time-Horizon Modeling Realities:** MCMC simulations demonstrate the necessity of utilizing a closed-population model (setting vital dynamics $\mu = 0$) to accurately map short-term, transient outbreak peaks, yielding an implied R_0 of 2.79 for the modeled initial surge. Conversely, long-term endemic analysis requires the inclusion of vital dynamics to prevent the susceptible population from artificially depleting.
- **Inflow Dynamics:** The hospitalization rate (p) acts as the primary accelerator dictating how rapidly the healthcare system hits its capacity ceiling, which subsequently triggers the public behavioral feedback loop.

Ultimately, this mathematical framework provides public health officials with a robust tool to evaluate the resilience of healthcare systems. It highlights that managing an epidemic requires not just expanding baseline hospital capacity, but also understanding the exact thresholds at which public awareness stabilizes infection dynamics.

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